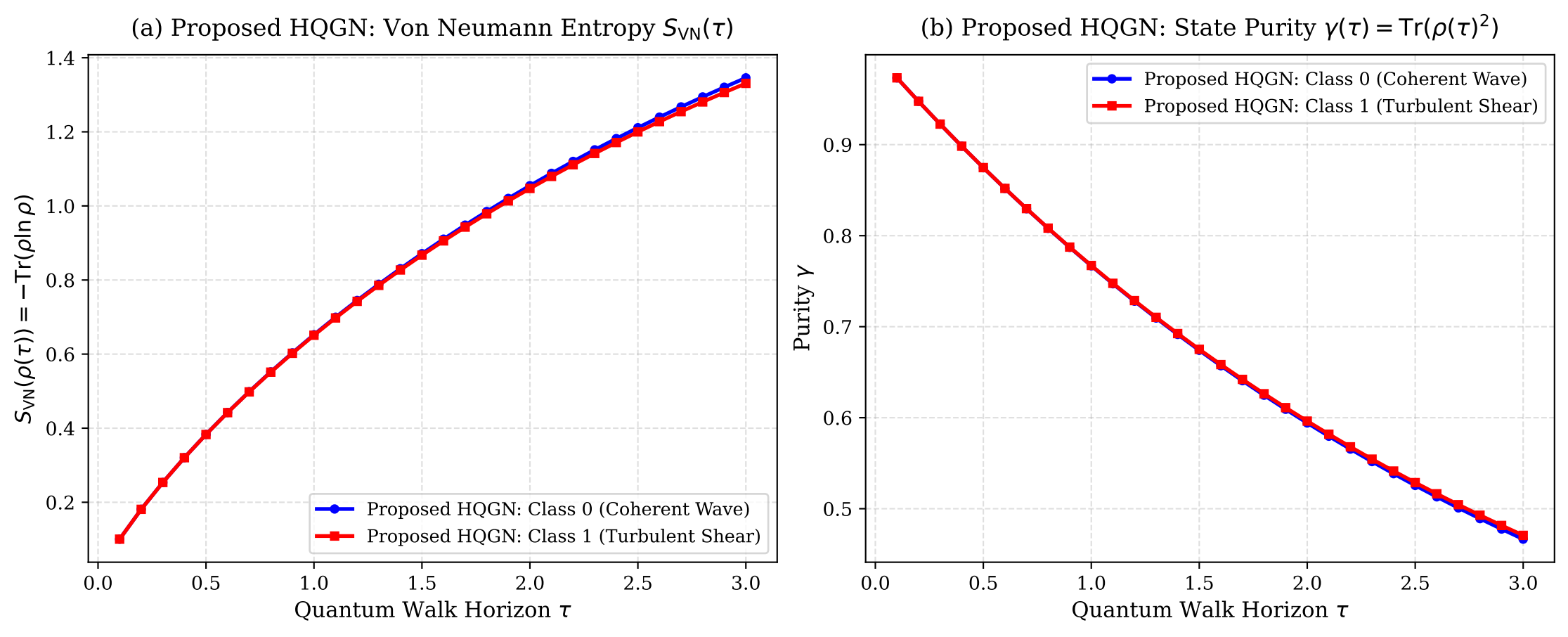
(a) Proposed HQGN: Von Neumann Entropy $S_{\mathrm{VN}}(\tau)$
$S_{\mathrm{VN}}(\rho(\tau)) = -\mathrm{Tr}(\rho \ln \rho)$
Quantum Walk Horizon τ
Proposed HQGN: Class 0 (Coherent Wave)
Proposed HQGN: Class 1 (Turbulent Shear)
(b) Proposed HQGN: State Purity $\gamma(\tau) = \mathrm{Tr}(\rho(\tau)^2)$
Purity γ
Quantum Walk Horizon τ
Proposed HQGN: Class 0 (Coherent Wave)
Proposed HQGN: Class 1 (Turbulent Shear)